

Limits

ANSWER KEY



Direct substitution

- 1 Find $\lim_{x \rightarrow 3} (2x^2 - x + 1)$.
 $= 2(9) - 3 + 1 = 16$.
- 2 Find $\lim_{x \rightarrow -1} (x^3 + 4x)$.
 $= (-1)^3 + 4(-1) = -1 - 4 = -5$.

Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ (for $x \neq 4$), so the limit is $4 + 4 = 8$.
- 4 Find $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 5x + 6}{x + 2} = \frac{(x + 2)(x + 3)}{x + 2} = x + 3$ (for $x \neq -2$), so the limit is $-2 + 3 = 1$.
- 5 Explain why you cannot find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.
Substituting $x = 4$ gives $\frac{16 - 16}{4 - 4} = \frac{0}{0}$, an indeterminate form — it does not tell you the limit is 0 or undefined, so you must factor and cancel first.

Limits at infinity for rational functions

- 6 Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.
The numerator and denominator have the same degree (2), so by Theorem 6.1 the limit is the ratio of leading coefficients: $\frac{3}{5}$.
- 7 Find $\lim_{x \rightarrow \infty} \frac{4x + 1}{2x^2 + 3}$.
The denominator has higher degree than the numerator, so by Theorem 6.1 the limit is 0.
- 8 Explain what happens to $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.
The numerator's degree (3) is higher than the denominator's (2), so by dividing by the highest power in the denominator the numerator's leading term still grows without bound — the limit does not exist as a finite number; the function increases without bound as $x \rightarrow \infty$.

Multiple choice — easy

9 Find $\lim_{x \rightarrow -7} (3x^2)$.

a. 145

b. 147 ✓

c. -21

d. 149

$$= 3(-7)^2 = 147.$$

10 Find $\lim_{x \rightarrow 7} (4x^2 + x - 7)$.

a. 194

b. 28

c. 198

d. 196 ✓

$$= 4(7)^2 + 7 - 7 = 196.$$

11 Find $\lim_{x \rightarrow -6} (2x^2 + 5x - 9)$.

a. 35

b. 31

c. -51

d. 33 ✓

$$= 2(-6)^2 - 30 - 9 = 33.$$

12 Find $\lim_{x \rightarrow -2} (-4x^2 + 8x - 3)$.

a. -35 ✓

b. -33

c. -11

d. -37

$$= -4(-2)^2 - 16 - 3 = -35.$$

13 Find $\lim_{x \rightarrow 2} (-3x^2 + 7x - 7)$.

a. -3

b. -5 ✓

c. -7

d. 1

$$= -3(2)^2 + 14 - 7 = -5.$$

14 Find $\lim_{x \rightarrow -7} (4x^2 - 2x - 9)$.

a. 203

b. 201 ✓

c. 199

d. -23

$$= 4(-7)^2 + 14 - 9 = 201.$$

15 Find $\lim_{x \rightarrow 5} (-3x^2 - 9x + 7)$.

a. -53

b. -113 ✓

c. -115

d. -111

$$= -3(5)^2 - 45 + 7 = -113.$$

16 Find $\lim_{x \rightarrow 5} (4x^2 + 4x - 4)$.

a. 116 ✓

b. 114

c. 118

d. 36

$$= 4(5)^2 + 20 - 4 = 116.$$

17 Find $\lim_{x \rightarrow -4} (3x^2 + x + 4)$.

a. -12

b. 50

c. 46

d. 48 ✓

$$= 3(-4)^2 - 4 + 4 = 48.$$

18 Find $\lim_{x \rightarrow -8} (3x^2 + 5x - 3)$.

a. 147

b. -67

c. 149 ✓

d. 151

$$= 3(-8)^2 - 40 - 3 = 149.$$

19 Find $\lim_{x \rightarrow 4} (3x^2 - 9x - 1)$.

a. -25

b. 11 ✓

c. 13

d. 9

$$= 3(4)^2 - 36 - 1 = 11.$$

20 Find $\lim_{x \rightarrow 6} (2x^2 + 9x - 6)$.

a. 122

b. 120 ✓

c. 60

d. 118

$$= 2(6)^2 + 54 - 6 = 120.$$

21 Find $\lim_{x \rightarrow -5} (-4x^2 + 2x - 2)$.

a. -112 ✓

b. -110

c. 8

d. -114

$$= -4(-5)^2 - 10 - 2 = -112.$$

22 Find $\lim_{x \rightarrow -4} (-2x^2 - 4x + 4)$.

a. -12 ✓

b. -10

c. -14

d. 28

$$= -2(-4)^2 + 16 + 4 = -12.$$

23 Find $\lim_{x \rightarrow -3} (3x^2 - 8x - 2)$.

a. 13

b. 51

c. 49 ✓

d. 47

$$= 3(-3)^2 + 24 - 2 = 49.$$

Multiple choice — medium

24 Find $\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{x - 3}$.

a. 1 ✓

b. 3

c. -1

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 5x + 6}{x - 3} = \frac{(x - 3)(x - 2)}{x - 3} = x - 2$ (for $x \neq 3$), so the limit is $3 - 2 = 1$.

25 Find $\lim_{x \rightarrow 8} \frac{x^2 - 17x + 72}{x - 8}$.

a. -3

b. 1

c. 2

d. -1 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 17x + 72}{x - 8} = \frac{(x - 8)(x - 9)}{x - 8} = x - 9$ (for $x \neq 8$), so the limit is $8 - 9 = -1$.

26 Find $\lim_{x \rightarrow -5} \frac{x^2 - 4x - 45}{x + 5}$.

- a. 14
- b. -16
- c. -12
- d. -14 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 4x - 45}{x + 5} = \frac{(x + 5)(x - 9)}{x + 5} = x - 9$ (for $x \neq -5$), so the limit is $-5 - 9 = -14$.

27 Find $\lim_{x \rightarrow 6} \frac{x^2 - 36}{x - 6}$.

- a. -12
- b. 10
- c. 12 ✓
- d. 14

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 36}{x - 6} = \frac{(x - 6)(x + 6)}{x - 6} = x + 6$ (for $x \neq 6$), so the limit is $6 + 6 = 12$.

28 Find $\lim_{x \rightarrow 4} \frac{x^2 - 9x + 20}{x - 4}$.

- a. -1 ✓
- b. 2
- c. -3
- d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 9x + 20}{x - 4} = \frac{(x - 4)(x - 5)}{x - 4} = x - 5$ (for $x \neq 4$), so the limit is $4 - 5 = -1$.

29 Find $\lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x + 2}$.

a. 2

b. -1 ✓

c. -3

d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 3x + 2}{x + 2} = \frac{(x + 2)(x + 1)}{x + 2} = x + 1$ (for $x \neq -2$), so the limit is $-2 + 1 = -1$.

30 Find $\lim_{x \rightarrow -2} \frac{x^2 + 2x}{x + 2}$.

a. -2 ✓

b. 2

c. -4

d. 0

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 2x}{x + 2} = \frac{(x + 2)(x)}{x + 2} = x$ (for $x \neq -2$), so the limit is $-2 = -2$.

31 Find $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$.

a. 7

b. 5 ✓

c. 3

d. -5

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - x - 6}{x - 3} = \frac{(x - 3)(x + 2)}{x - 3} = x + 2$ (for $x \neq 3$), so the limit is $3 + 2 = 5$.

32 Find $\lim_{x \rightarrow -1} \frac{x^2 - 4x - 5}{x + 1}$.

a. -6 ✓

b. -8

c. 6

d. -4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 4x - 5}{x + 1} = \frac{(x + 1)(x - 5)}{x + 1} = x - 5$ (for $x \neq -1$), so the limit is $-1 - 5 = -6$.

33 Find $\lim_{x \rightarrow 5} \frac{x^2 - 5x}{x - 5}$.

a. -5

b. 5 ✓

c. 3

d. 7

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 5x}{x - 5} = \frac{(x - 5)(x)}{x - 5} = x$ (for $x \neq 5$), so the limit is $5 = 5$.

34 Find $\lim_{x \rightarrow 6} \frac{x^2 - 2x - 24}{x - 6}$.

a. 8

b. -10

c. 12

d. 10 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 2x - 24}{x - 6} = \frac{(x - 6)(x + 4)}{x - 6} = x + 4$ (for $x \neq 6$), so the limit is $6 + 4 = 10$.

35 Find $\lim_{x \rightarrow 7} \frac{x^2 - 16x + 63}{x - 7}$.

a. -2 ✓

b. -4

c. 0

d. 2

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 16x + 63}{x - 7} = \frac{(x - 7)(x - 9)}{x - 7} = x - 9$ (for $x \neq 7$), so the limit is $7 - 9 = -2$.

36 Find $\lim_{x \rightarrow -6} \frac{x^2 + 2x - 24}{x + 6}$.

a. -10 ✓

b. -8

c. -12

d. 10

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 2x - 24}{x + 6} = \frac{(x + 6)(x - 4)}{x + 6} = x - 4$ (for $x \neq -6$), so the limit is $-6 - 4 = -10$.

37 Find $\lim_{x \rightarrow -5} \frac{x^2 + 9x + 20}{x + 5}$.

a. -3

b. -1 ✓

c. 2

d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 9x + 20}{x + 5} = \frac{(x + 5)(x + 4)}{x + 5} = x + 4$ (for $x \neq -5$), so the limit is $-5 + 4 = -1$.

38 Find $\lim_{x \rightarrow -9} \frac{x^2 + 14x + 45}{x + 9}$.

- a. -2
- b. 4
- c. -4 ✓
- d. -6

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 14x + 45}{x + 9} = \frac{(x + 9)(x + 5)}{x + 9} = x + 5$ (for $x \neq -9$), so the limit is $-9 + 5 = -4$.

Multiple choice — hard

39 Find $\lim_{x \rightarrow \infty} \frac{-x^2 + 3x}{3x^2 + x}$.

- a. $\frac{1}{3}$
- b. $\frac{2}{3}$
- c. $-\frac{4}{3}$
- d. $-\frac{1}{3}$ ✓

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients:
 $\frac{-1}{3} = -\frac{1}{3}$.

40 Find $\lim_{x \rightarrow \infty} \frac{-6x^2 + 6x}{8x^2}$.

- a. $-\frac{7}{4}$
- b. $\frac{3}{4}$
- c. $-\frac{3}{4}$ ✓
- d. $\frac{1}{4}$

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients:
 $\frac{-6}{8} = -\frac{3}{4}$.

- 41 Find $\lim_{x \rightarrow \infty} \frac{-7x^2 + 3x}{-5x^2 + 4x}$.
- a. $\frac{7}{5}$ ✓
- b. $-\frac{7}{5}$
- c. $\frac{2}{5}$
- d. $\frac{12}{5}$

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients:
 $\frac{-7}{-5} = \frac{7}{5}$.

- 42 Find $\lim_{x \rightarrow \infty} \frac{-6x^2 + x}{-4x^2 - 4x}$.
- a. $\frac{3}{2}$ ✓
- b. $-\frac{3}{2}$
- c. $\frac{5}{2}$
- d. $\frac{1}{2}$

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients:
 $\frac{-6}{-4} = \frac{3}{2}$.

- 43 Find $\lim_{x \rightarrow \infty} \frac{-8x^2 + 2x}{9x^2 + x}$.
- a. $-\frac{17}{9}$
- b. $\frac{1}{9}$
- c. $-\frac{8}{9}$ ✓
- d. $\frac{8}{9}$

The numerator and denominator have the same degree (2), so the limit is the ratio of leading coefficients:
 $\frac{-8}{9} = -\frac{8}{9}$.

- 44 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+81} - \sqrt{81}}{x}$.
- a. $\frac{1}{18}$ ✓
- b. $\frac{1}{9}$
- c. $\frac{1}{6}$
- d. $\frac{2}{9}$

Multiply top and bottom by the conjugate $\sqrt{x+81} + \sqrt{81}$: the numerator becomes x , which cancels with the x in the denominator, leaving $1/(\sqrt{x+81} + \sqrt{81})$. As $x \rightarrow 0$: this is $1/(\sqrt{81} + \sqrt{81}) = 1/(2\sqrt{81}) = 1/(2 \cdot 9) = \frac{1}{18}$.

- 45 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+64} - \sqrt{64}}{x}$.
- a. $\frac{1}{4}$
- b. $\frac{1}{8}$
- c. $\frac{1}{16}$ ✓
- d. $\frac{3}{16}$

Multiply top and bottom by the conjugate $\sqrt{x+64} + \sqrt{64}$: the numerator becomes x , which cancels with the x in the denominator, leaving $1/(\sqrt{x+64} + \sqrt{64})$. As $x \rightarrow 0$: this is $1/(\sqrt{64} + \sqrt{64}) = 1/(2\sqrt{64}) = 1/(2 \cdot 8) = \frac{1}{16}$.

- 46 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+144} - \sqrt{144}}{x}$.
- a. $\frac{1}{6}$
- b. $\frac{1}{8}$
- c. $\frac{1}{24}$ ✓
- d. $\frac{1}{12}$

Multiply top and bottom by the conjugate $\sqrt{x+144} + \sqrt{144}$: the numerator becomes x , which cancels with the x in the denominator, leaving $1/(\sqrt{x+144} + \sqrt{144})$. As $x \rightarrow 0$: this is $1/(\sqrt{144} + \sqrt{144}) = 1/(2\sqrt{144}) = 1/(2 \cdot 12) = \frac{1}{24}$.

- 47 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+25} - \sqrt{25}}{x}$.
- a. $\frac{2}{5}$
 - b. $\frac{3}{10}$
 - c. $\frac{1}{10}$ ✓
 - d. $\frac{1}{5}$

Multiply top and bottom by the conjugate $\sqrt{x+25} + \sqrt{25}$: the numerator becomes x , which cancels with the x in the denominator, leaving $1/(\sqrt{x+25} + \sqrt{25})$. As $x \rightarrow 0$: this is $1/(\sqrt{25} + \sqrt{25}) = 1/(2\sqrt{25}) = 1/(2 \cdot 5) = \frac{1}{10}$.

- 48 Find $\lim_{x \rightarrow 0} \frac{\sqrt{x+49} - \sqrt{49}}{x}$.
- a. $\frac{2}{7}$
 - b. $\frac{3}{14}$
 - c. $\frac{1}{14}$ ✓
 - d. $\frac{1}{7}$

Multiply top and bottom by the conjugate $\sqrt{x+49} + \sqrt{49}$: the numerator becomes x , which cancels with the x in the denominator, leaving $1/(\sqrt{x+49} + \sqrt{49})$. As $x \rightarrow 0$: this is $1/(\sqrt{49} + \sqrt{49}) = 1/(2\sqrt{49}) = 1/(2 \cdot 7) = \frac{1}{14}$.

- 49 Find $\lim_{x \rightarrow \infty} \frac{x^3 + 5x}{9x^2 - 9x}$ (state whether it is ∞ or $-\infty$).
- a. $\frac{1}{9}$
 - b. 0
 - c. $-\infty$
 - d. ∞ ✓

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{1}{9}x$ for large x . Since the leading coefficients 1 and 9 have same sign, the limit is ∞ .

50 Find $\lim_{x \rightarrow \infty} \frac{-9x^3 - 8x}{-8x^2 + 2x}$ (state whether it is ∞ or $-\infty$).

a. ∞ ✓

b. $\frac{9}{8}$

c. $-\infty$

d. 0

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{-9}{-8}x$ for large x . Since the leading coefficients -9 and -8 have same sign, the limit is ∞ .

51 Find $\lim_{x \rightarrow \infty} \frac{-2x^3 - x}{3x^2 + 9x}$ (state whether it is ∞ or $-\infty$).

a. $-\frac{2}{3}$

b. 0

c. ∞

d. $-\infty$ ✓

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{-2}{3}x$ for large x . Since the leading coefficients -2 and 3 have opposite sign, the limit is $-\infty$.

52 Find $\lim_{x \rightarrow \infty} \frac{6x^3 + 7x}{2x^2 + 6x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. 3

c. ∞ ✓

d. $-\infty$

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{6}{2}x$ for large x . Since the leading coefficients 6 and 2 have same sign, the limit is ∞ .

53 Find $\lim_{x \rightarrow \infty} \frac{2x^3 - 8x}{9x^2 - 3x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. $\frac{2}{9}$

c. $-\infty$

d. ∞ ✓

The numerator's degree (3) is higher than the denominator's (2), so the quotient behaves like $\frac{2}{9}x$ for large x . Since the leading coefficients 2 and 9 have same sign, the limit is ∞ .